

Frontend Engineering Fundamentals: A 6-Week Intensive Course

WEEK 1: HTML & CSS Foundations

Learning Objectives

By the end of this week, students will:

- Understand HTML document structure and semantic markup
- Write clean, accessible HTML
- Style web pages with CSS including layouts, typography, and colors
- Use CSS Flexbox for responsive layouts
- Understand the box model and positioning

Learning Materials

HTML Fundamentals

Videos:

- "HTML Tutorial for Beginners" by Programming with Mosh (1 hour)
<https://www.youtube.com/watch?v=qz0aGYrrlhU>
- "HTML Semantic Elements" by Web Dev Simplified (20 mins)
https://www.youtube.com/watch?v=kGW8Al_cga4

Reading:

- MDN Web Docs - HTML Basics
https://developer.mozilla.org/en-US/docs/Learn/Getting_started_with_the_web/HTML
- MDN - Structuring Content with HTML
https://developer.mozilla.org/en-US/docs/Learn/HTML/Introduction_to_HTML

Github

- GitHub Guides - Hello World

<https://guides.github.com/activities/hello-world/>

Practice (Test your knowledge): Build a simple personal bio page with proper semantic HTML

CSS Fundamentals

Videos:

- "CSS Tutorial - Zero to Hero" by freeCodeCamp (11 hours - watch sections 1-4)

<https://www.youtube.com/watch?v=1Rs2ND1ryYc>

- "Learn Flexbox in 15 Minutes" by Web Dev Simplified

<https://www.youtube.com/watch?v=fYq5PXgSsbE>

Reading:

- CSS-Tricks - A Complete Guide to Flexbox

<https://css-tricks.com/snippets/css/a-guide-to-flexbox/>

- MDN - CSS Basics

https://developer.mozilla.org/en-US/docs/Learn_web_development/Getting_started/Your_first_website/Styling_the_content

Practice: Style your bio page with colors, fonts, and basic layout

Advanced CSS & Layouts

Videos:

- "Build a Responsive Website" by Traversy Media (1.5 hours)

<https://www.youtube.com/watch?v=p0bGHP-PXD4>

- "CSS Units Explained" by Kevin Powell

https://www.youtube.com/watch?v=N5wpD9Ov_To

Reading:

- MDN - CSS Box Model

https://developer.mozilla.org/en-US/docs/Learn/CSS/Building_blocks/The_box_model

- MDN - CSS Positioning

<https://developer.mozilla.org/en-US/docs/Web/CSS/position>

Practice: Create responsive navigation and footer

Week 1 Challenge Project

Build a Multi-Page Portfolio Website

Requirements:

- Create 3 pages: Home, About, and Projects (with placeholder content)
- Must include: navigation bar, hero section, contact form (non-functional), footer
- Fully responsive (mobile, tablet, desktop)
- Use semantic HTML and demonstrate Flexbox mastery

Deliverable: Live site link (GitHub Pages) + code repository

Success Criteria:

- Clean, semantic HTML structure
- Responsive design that works on all screen sizes
- Consistent styling and spacing
- Proper use of Flexbox for layouts
- Accessible markup (alt tags, proper heading hierarchy)